PARTNER PRODUCT: MKBANK, CREDIT MONITORING DEPARTMENT

Maqsad

Remotely verifies that a subsidised loan was spent on its stated purpose, using the money trail, photos and invoices. A bank officer makes the decision.

Live demo

FinTech / AI in Finance · WIUT Hackathon 2026 · Team Maqsad

PROBLEM

Part of subsidised lending never reaches its purpose

68 bn

UZS of subsidised loans misused in 2,500 cases found by regional inspections

~3 trn

UZS of family-business programme loans still not repaid — misuse is one of the drivers

Monitoring is the bank's duty. Today an officer visits each borrower in person — and borrowers are many and scattered

Sources: anhor.uz (meeting with bank heads); yuz.uz; Presidential Resolution PQ-5041

STATUS QUO AND ALTERNATIVES

Today's tools don't check the purpose

	On-site visit today's process	AML monitoring existing bank tools	**Maqsad** remote, evidence-based
Checks the loan's purpose	✓ Yes — by eye, on site	⚠ No — looks for money laundering	✓ Yes — money trail, photo and invoice
Reaches scattered borrowers	⚠ Slow — one trip per borrower	✓ Yes — remote	✓ Yes — remote, ranked by risk
Cost per check	⚠ 100–300k UZS (assumed)	Not applicable	✓ ~15k UZS
Evidence behind each decision	Officer's notes	Laundering alerts	✓ Each reason linked to evidence, logged

SOLUTION

Three signals, one queue, a human decision

Money trail — the core signal

Cashed out on day one? Supplier paid? The bank's own data

Asset photo

Matches the purpose? Not a duplicate? Place and time

Invoice

Does the supplier's business code (OKED) fit the purpose?

Risk queue

- **Red** — check first
- **Yellow** — watch
- **Green** — remote check is enough

The **officer** approves, rejects or schedules a visit — every decision is logged

LIVE DEMO

Try it yourself — scan with your phone



1 • Risk queue

Open a red loan — see why it is red



2 • Photo check

Pick the sewing-machine loan, photograph any object



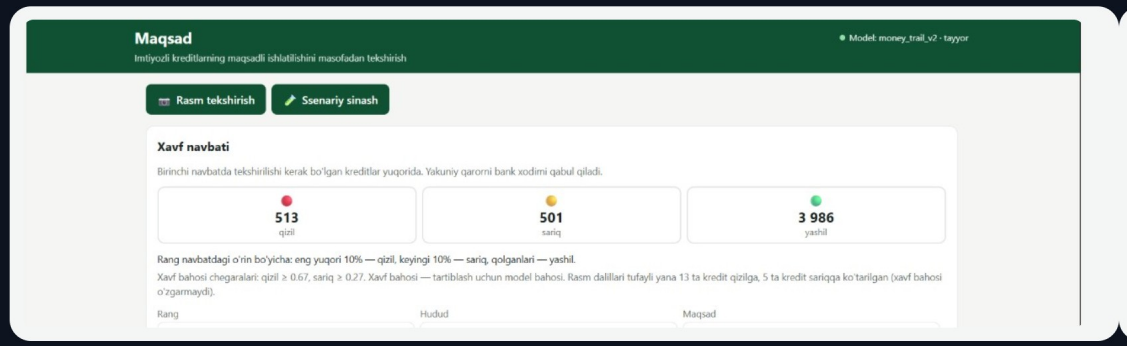
3 • Scenario

Enter your own money flows, get a risk score

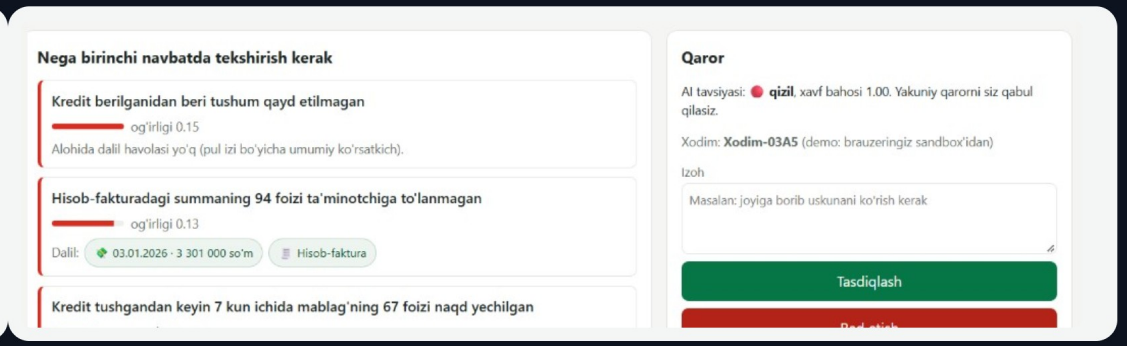
maqsad-production-d704.up.railway.app · every visitor gets a private sandbox — your results are visible only to you

PRODUCT · DEMO BACKUP

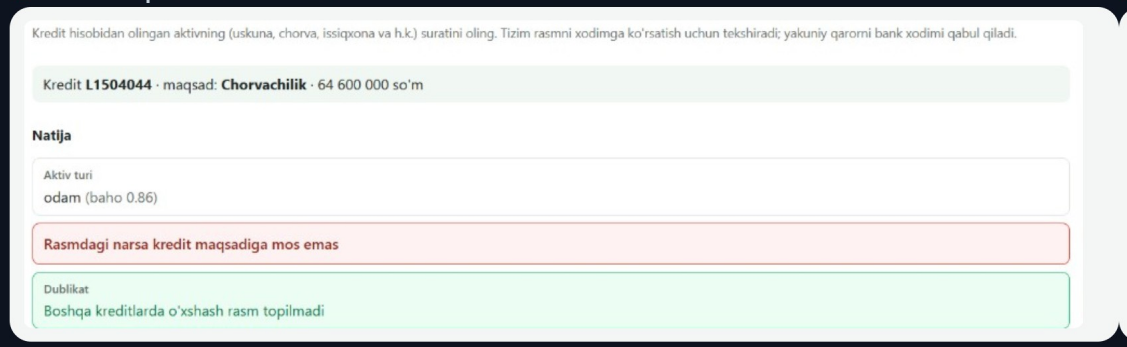
Four screens the officer sees



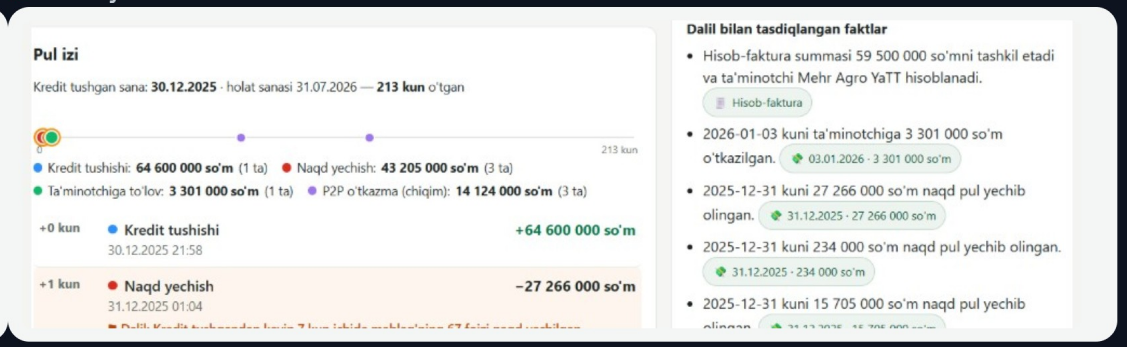
1 · Risk queue — red loans come first



2 · Why red: each reason linked to evidence



3 · Photo check: does not match the purpose



4 · Money trail and draft report

RESULTS · MODELS WE BUILT, QUALITY WE MEASURED

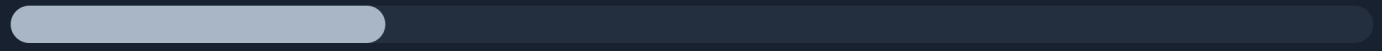
Same budget, almost 3× the cases caught

Fraud cases caught when officers visit only **20% of loans**

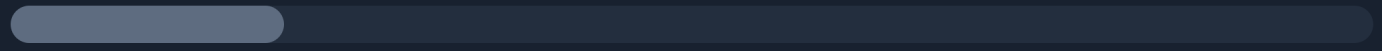
Maqsad — money-trail model **78.2%**



Simple rule — same budget **27.4%**



Random selection — same budget **20%**



2.9× the simple rule · 3.9× random selection

PHOTO CHECK

94.8%

asset-type accuracy on 139 open-licence photos (CLIP + our linear probe)

Zero-shot CLIP alone: 89.9%

Money trail: LightGBM with plain-language reasons · Photo: duplicate search, asset type, place and time, faces blurred · LLM only drafts the report; every fact is checked in code

78.2% and 27.4%: synthetic hold-out set, separate seed; random selection catches 20% by definition. 94.8%: real photos, cross-validation grouped by photographer, measured after removing personal data.

DEPLOYMENT, AUDIT AND COMPLIANCE

Customer data never leaves the bank

Bank server (on-prem)

ML modules	App server	Local LLM
Bank data	Audit log · hash chain	

Measured on the live prototype	Value
Photo check, p50	363 ms
Queue (50 loans), p50	292 ms
Draft report, n=3	3.9–6.0 s
One draft report	$0.0034

Prototype LLM: Gemini API. Production: a local model — same code, only LLM_BASE_URL changes

Explainable

Every flag has plain-language reasons linked to evidence

Auditable

Who, when, AI advice, officer decision, model version — change one record and the chain breaks

Human in the loop

AI only ranks the queue; the officer decides every case

BUILT-IN SAFEGUARDS

Known risks — and how we handle them

Synthetic money trail

Patterns taken from real reported cases; pilot month 1 measures accuracy on the bank's historical loans

A photo can be staged

It never decides alone: duplicate search, place and time — and the money trail outweighs it

The model can miss a case

It only ranks the queue; an officer still decides and visits every red loan

The LLM can make things up

Every fact is checked against evidence in code; unverified facts are flagged, never hidden

45%
caught

Weakest case we measured: a fake invoice through a correctly registered supplier

That is why the photo check and the officer stay in the loop.

BUSINESS CASE · ONE BRANCH, 1,540 LOANS

Remote checks pay for themselves fast

Base case

~3 mo

payback

Half the visits stay physical

~5 mo

payback

All inputs at worst case

~34 mo

payback

Unit cost per check

~15k UZS

remote check with Maqsad

100–300k UZS

on-site visit (assumed)

~7–20× cheaper per check

Business model

Annual licence per branch

Priced below the visits it saves. Recovered losses are extra upside.

Visit savings only; recovered losses are not included — measured in pilot month 1. Some inputs are assumptions. Source: ROI model v3.3

GO-TO-MARKET · PILOT PLAN

One branch, three months, measurable results

Month 1

Install and test

Install on the bank server, test on historical loans, measure how many cases today's method finds

Month 2

Run in parallel

Officers use the system alongside the current process; we tune it on their feedback

Month 3

Measure

Cases caught, visits saved and real accuracy — then the decision to scale

Go / no-go at month 3 on three numbers: Cases caught · Visits saved · Real accuracy

Existing AML tools look for money laundering, not for whether a loan was used for its purpose — so the pilot measures against today's visits, not against another tool.

TEAM

Three people, three layers

Dilshodbek Tolibjonov

AI / ML

Synthetic portfolio, money-trail model, photo analysis, evaluation

Oybek Xushnazarov

Application

API, officer dashboard, camera, LLM report, audit, deployment

Shohijahon Shavkatov

Business and pitch

Problem evidence, ROI model, pilot plan

</> Public code: github.com/dilshodbekio/maqsad

OUR ASK

A one-branch pilot with MKBANK

3 months · on the bank's server · the officer decides

1. One branch and its historical loans, on the bank's own server
2. Officers' time to run Maqsad alongside today's process
3. A month-3 review: cases caught, visits saved, real accuracy

78.2%
fraud caught visiting 20% of loans

~3 mo
payback in the base case

0
external APIs in production

Try the live site

Demo: maqsad-production-d704.up.railway.app · Code: github.com/dilshodbekio/maqsad · Synthetic money-trail data; real accuracy measured in the pilot